

2022 Second round Nepal Mathematics Olympiad

October 30, 2022

Problem 1

Determine all real solutions of equation:

$$\sqrt{x^2 - r} + 2\sqrt{x^2 - 1} = x, x \in R$$

Problem 2

Find all rational solutions (x, y) which satisfies the equation:

$$x^2 + 2y^2 = 4xy$$

Problem 3 An alien convoy of 4200 aliens from a different universe set out to study all the 2022 planets in an alternative universe. They are given following information:

- a) Each and every alien should travel all the planets without repetition.
- b) They are only allowed to travel a safe path, the number of safe path is given by the number of trailing zeroes in the binary representation of total number of possible paths.

For example: for 6 possible paths, it's binary representation is 110_2 which has only 1 trailing 0, so there is only 1 safe path. Prove that, no matter which path they choose, there should be at least 3 aliens that choose the same path.

Problem 4

Find the area of the shaded region in the following figure in which sides AB, BC and AC are 8, 7 and 9 units respectively.

